

IMPROVED INTERCONNECT TECHNOLOGIES FOR 3D ICs AND 3D PACKAGES

The electronics industry continues to strive towards a common goal – to pack more functionality into smaller form factors. For this, the industry depends on constant reductions in the size of chip packages. Maintaining the tempo predicted by the famous Moore's Law, the integrated circuit (IC) makers are constantly improving their fabrication processes and designs to fit more components into smaller footprints. However, the electronics industry is not satisfied with even that – they want more!

This has resulted in three-dimensional (3D) ICs and packages. A 3D IC is basically a stack of multiple silicon wafers and dies connected through vertical electrical connections, such as through-silicon vias (TSVs), in a way that they behave like a single device. Wikipedia defines a TSV as a "vertical electrical connection (via) passing completely through a silicon wafer or die."

A step above, we have 3D packages, that is, interconnected stacks of



two or more chips. The stacked chips may be wired together along their edges, but vertical connection using TSVs is now emerging as the more popular option since it does not add to the dimensions of the stack. There are several TSV designs and production technologies in use.

Of course, engineers are never satisfied – even as TSV is gaining popularity, next-generation technologies are already being demonstrated. One such wireless interconnect technology was described by Yasufumi Sugimori at the International Solid-State Circuit

Conference 2009. The technology used coupled inductors to send signals between stacked die, across a distance of 120 µm. This coupling avoids the need for TSVs, saving the cost of this wafer processing step. According to Sugimori's estimate, the power spent communicating through the stacked chips will only be half of what is spent in today's stacked chip packages, while the area spent on communication circuits can be reduced 40 times.

BIOELECTRONICS

This is the age of technological convergence – most significant breakthroughs happen at the interface of one or more fields of science. Such a conjunction of electrical and electronics engineering with biology, physics, chemistry and material sciences is going to be the future of healthcare. We are talking about such advanced technology that will make current equipment like cardiac pacemakers and blood glucose meters seem very simple!

'Wetware' comprising bioelectronics and biosensors will make it possible to measure and understand biological systems in greater detail and with unprecedented accuracy. This will result in advanced lab-on-a-chip biomedical diagnosis, implantable neural interfaces, the ability to restore a patient's lost vision or reverse the effects of spinal cord injuries, etc. An EETimes.com report cites: "Lab-on-a-chip is one manifestation of the technology... but it is also possible to grow biological cells on electronically addressable substrates. The opportunities for in-vitro diagnostics are clear. Information about the electrical behaviour of individual cells and their reactions to drugs is a major focus for research in cardiac and neural ailments such as Alzheimer's or Parkinson's disease."

Bioelectronics shows a lot of promise now, thanks to the rapid increase in the understanding of biological systems and processes at the nano-scale, advances in semiconductor technology and in surface chemistry related to the interface of biology and man-made devices, availability of micro energy sources, biosensors and such miniature measurement systems, and the materials and fabrication techniques for bio-electronic devices, medical



implants, 3D assembly, self-assembly, nanoparticles, nano-tubes, nano-wires, etc.

A report by the US National Institute of Standards and Technology (NIST) beautifully describes how it is not just a case of miniature electronics and MEMS enhancing medical treatments but also a case wherein the electronics industry learns from the ways and means of biological systems to enhance its own capabilities at the nano-scale. The 2009 report titled *A Framework for Bioelectronics: Discovery and Innovation*

states that: "understanding biology may provide powerful insights into efficient assembly processes, devices, and architectures for nano-electronics technology, as physical limits of existing technologies are approached... Advances in bioelectronics can offer new and improved methods and tools while simultaneously reducing their costs, due to the continuing exponential gains in functionality-per-unit-cost in nanoelectronics (aka Moore's Law). These gains drove the cost per transistor down by a factor of one million between 1970 and 2008 (for comparison, over the same period, the average cost of a new car rose from \$3,900 to \$26,000) and enabled unprecedented increases in productivity."

This win-win outlook will contribute further to the growth of bioelectronics. Several industry majors including IBM, Intel, Texas Instruments and Freescale are involved in the research and development of wetware. If there is as much development in the field as we expect to see, it will—over time, and as prices fall—lead to revolutionary changes in national security, healthcare and economic development too.